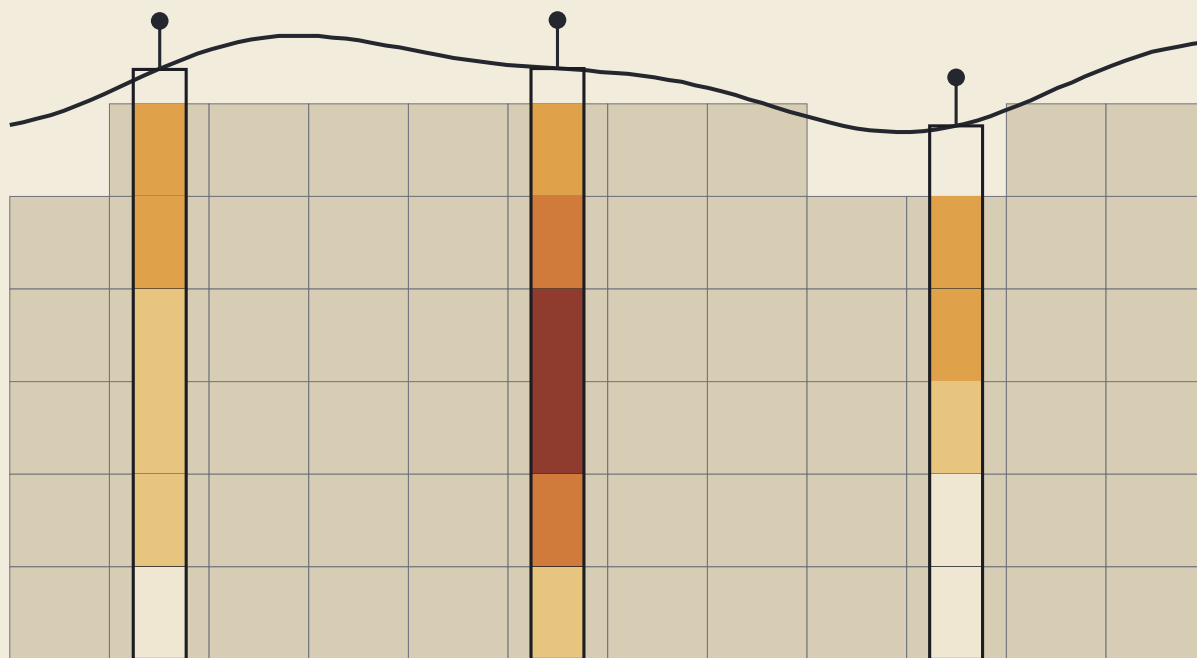




all of geostatistics, one sayable line at a time

WHAT'S UNDER THE HILL?

**For every kid who ever wondered
what's under the hill.**

Written and drawn by Arthur Endlein Correia.

First edition --- Belo Horizonte, 2026.

Geoscientific Chaos Union.

ISBN [to be assigned]

This book is free. The text and figures are released under CC0; the code is MIT. Copy it, print it, remix it, sell it.

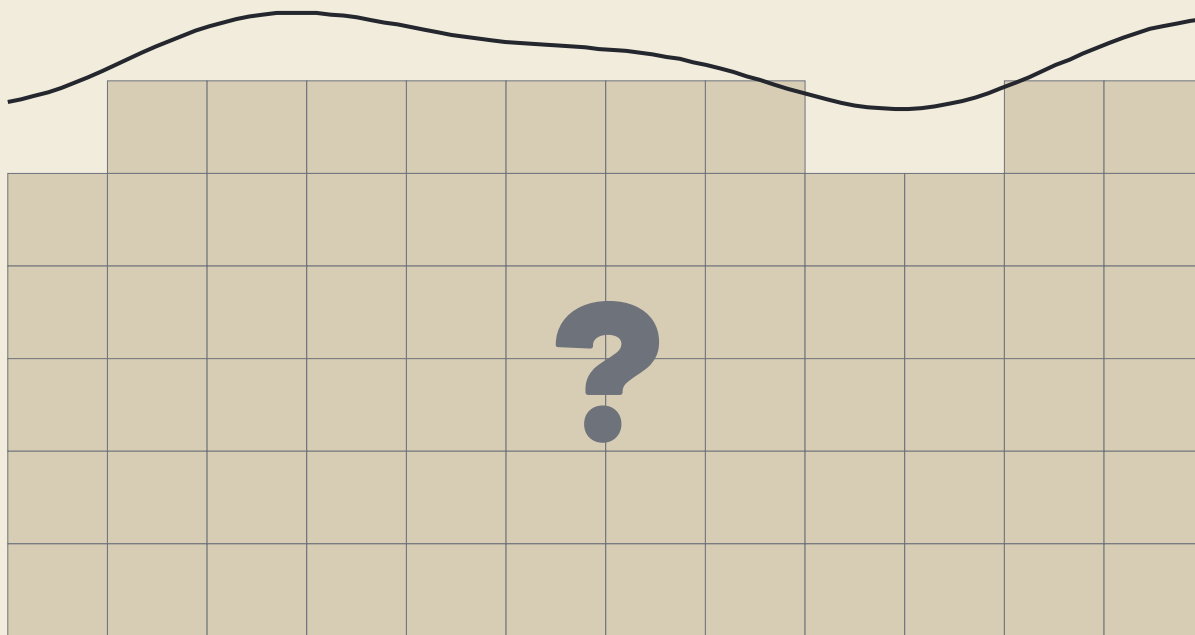
Set in Barlow and Space Mono (SIL Open Font License). Rebuilds, identically, from fieldkit_full.py and random seed 7.

[ficha catalográfica --- CBL]

ACT I

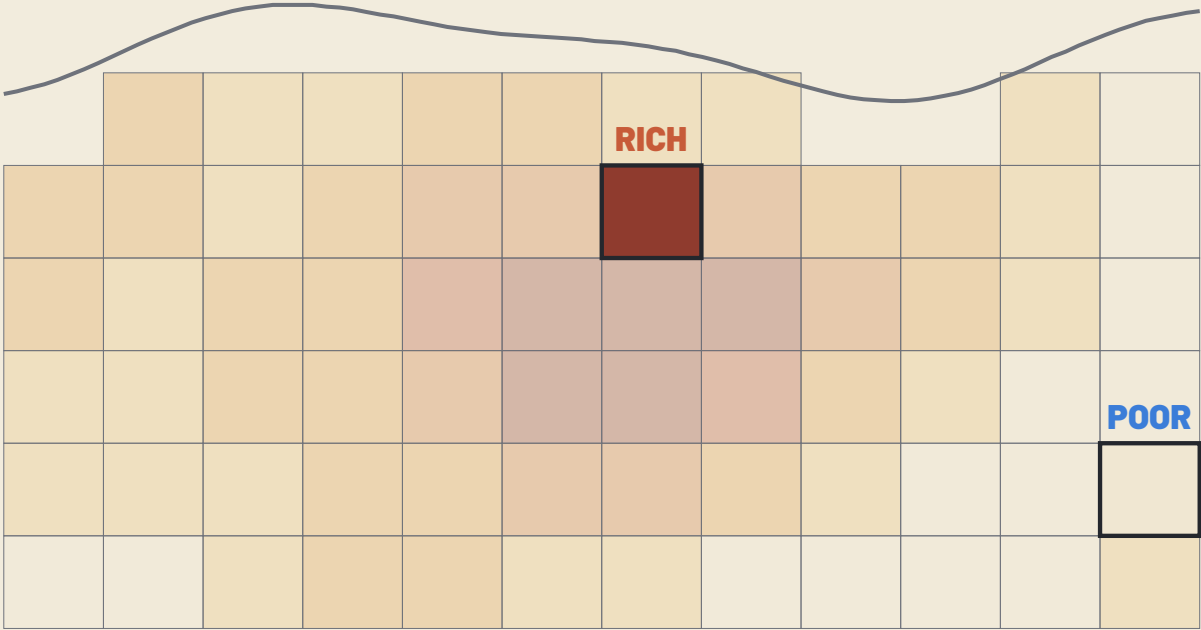
THE UNKNOWN

the problem, and the few clues we get



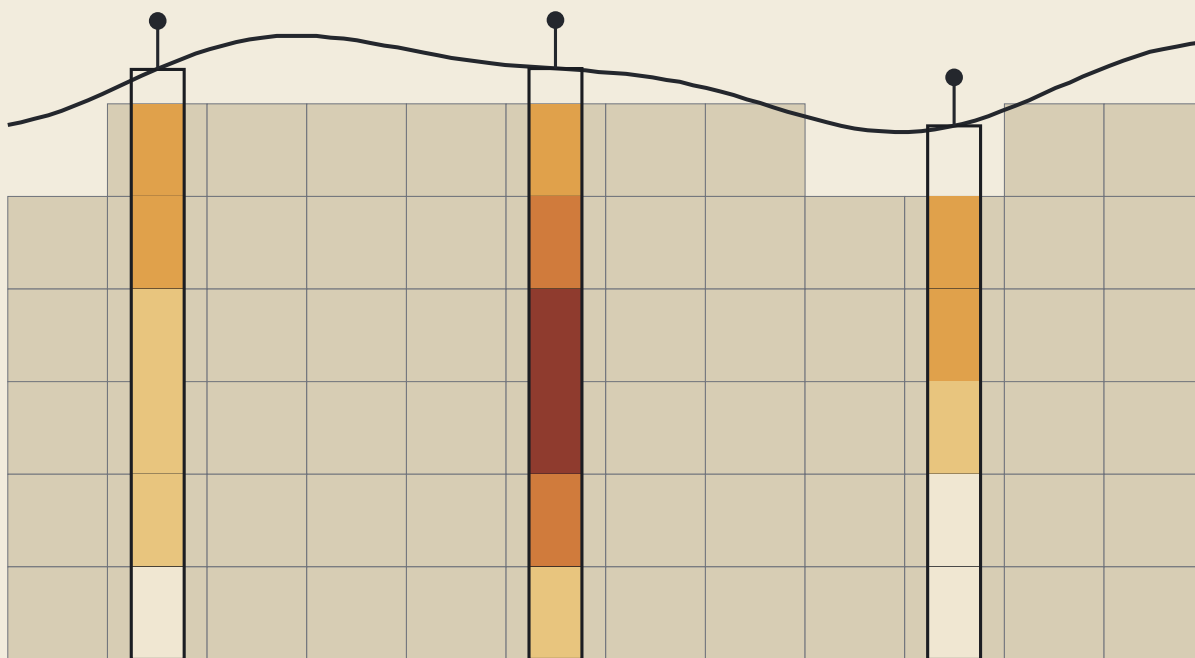
What's under the hill?
We can't see inside.

// the unknown --- a property that lives in space, $Z(x)$



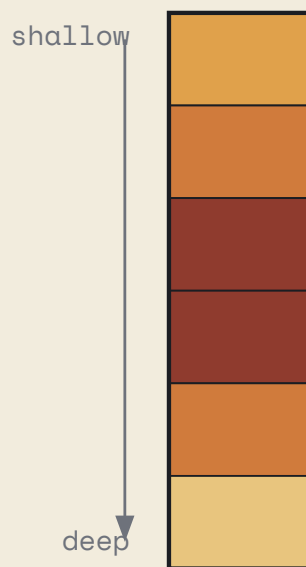
**Some rock is rich. Some is poor.
It changes from place to place.**

// spatial variability --- the value differs everywhere



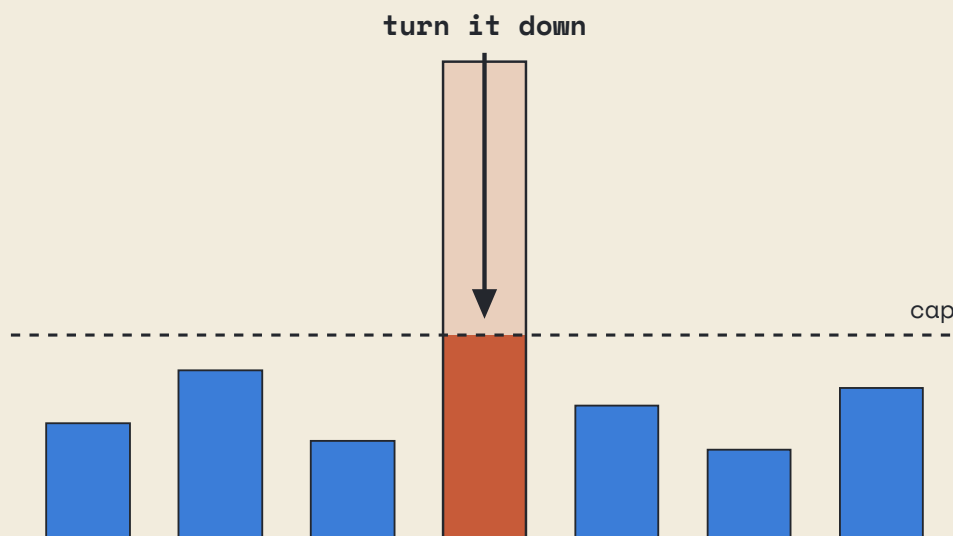
**We can't dig it all up.
So we drill a few holes.**

// sampling --- a sparse set of observations



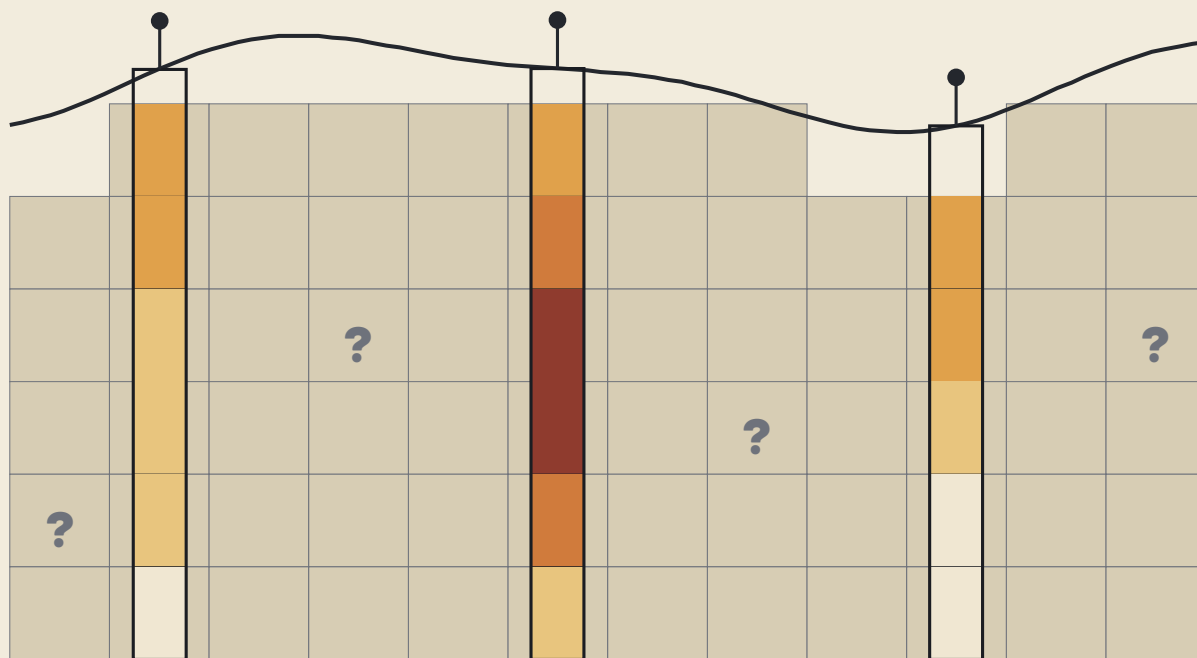
**Go down a hole --
the colours change with depth.**

// down-hole assays --- each interval of core is its own sample



**One rock shouts a giant number.
Do we believe it — or turn it down?**

// capping / top-cut --- taming an extreme value so one sample can't dominate



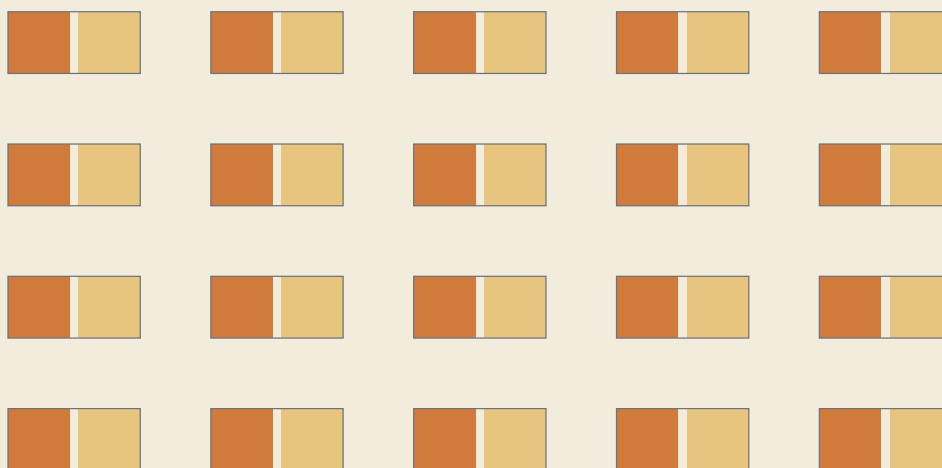
A few holes. A big hill.
Mostly a guess.

// the estimation problem --- much to predict, little data

ACT II

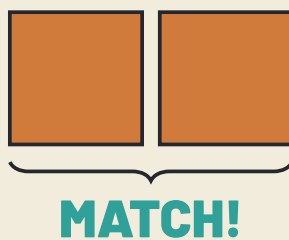
HOW THE GROUND HANGS TOGETHER

spatial correlation, and the curve that measures it



**We pretend the hill keeps
the same habits all over.**

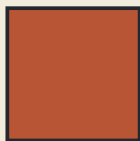
// stationarity --- the rules of variation are assumed the same everywhere



**Rocks close together
are alike.**

// spatial correlation --- Tobler's first law of geography

NOT SO MUCH



Rocks far apart?
Not so much.

// correlation fades as separation grows



still a tiny bit different

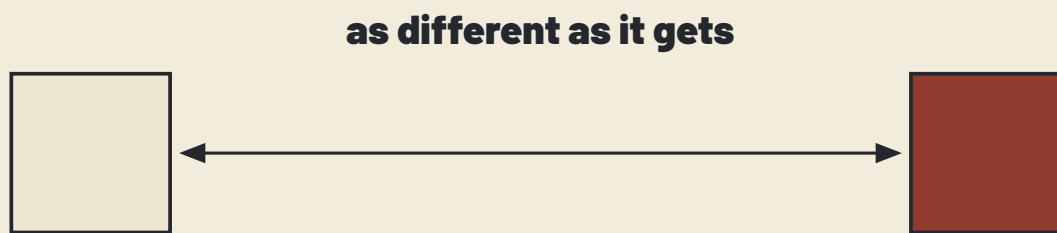
**Side by side,
they still aren't exactly the same.**

// the nugget --- variability that survives even at zero distance



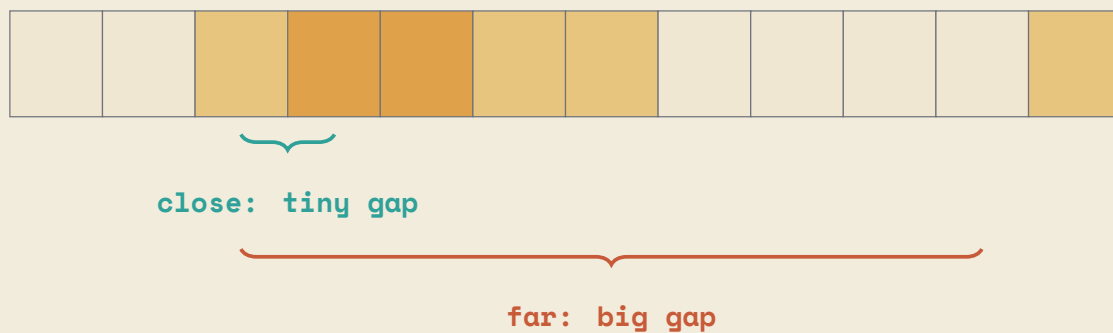
**Far enough apart,
rocks forget each other.**

// the range --- the distance beyond which there is no correlation



**All the way apart =
the biggest difference there is.**

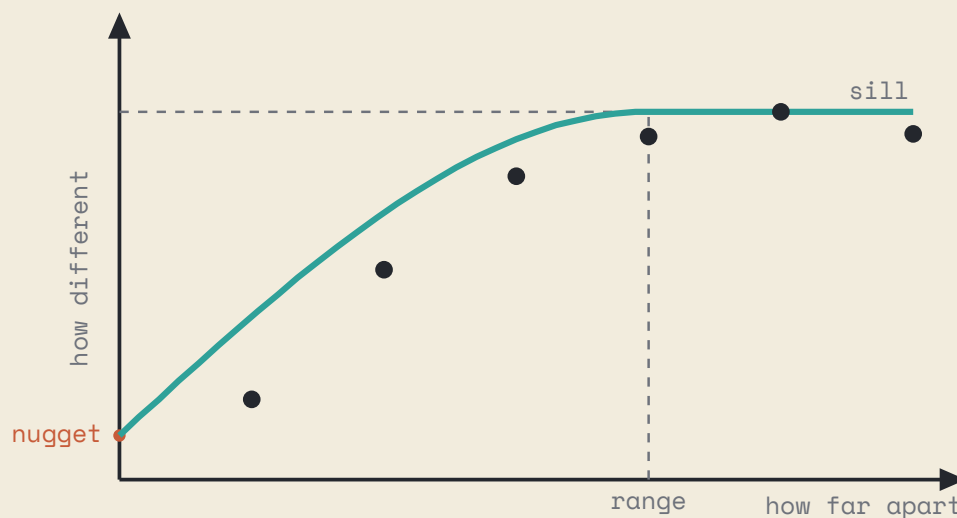
// the sill --- the total variance, reached at the range



Pick two rocks. How far? How different?
Again. And again.

// lag pairs --- the experimental variogram is built from every pair

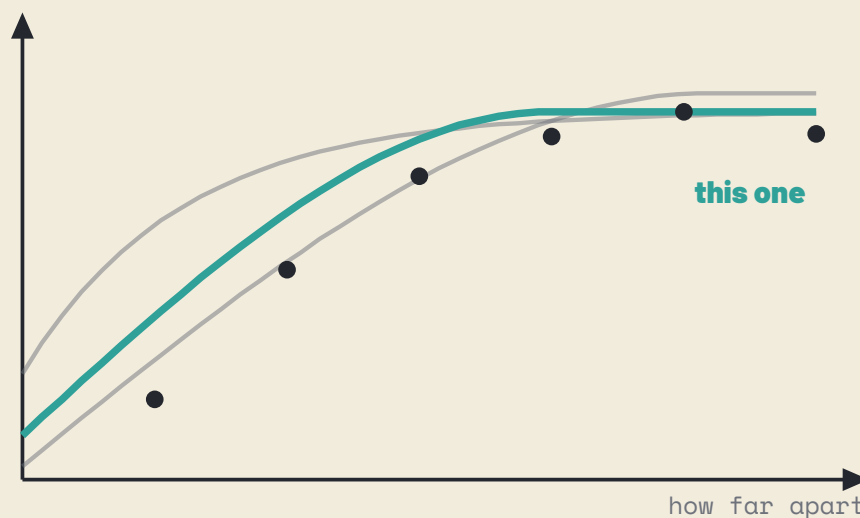
FIG.14 --- THE DIFFERENCE CURVE



**Close pairs barely differ. Far pairs differ a lot.
Draw it — that's the curve!**

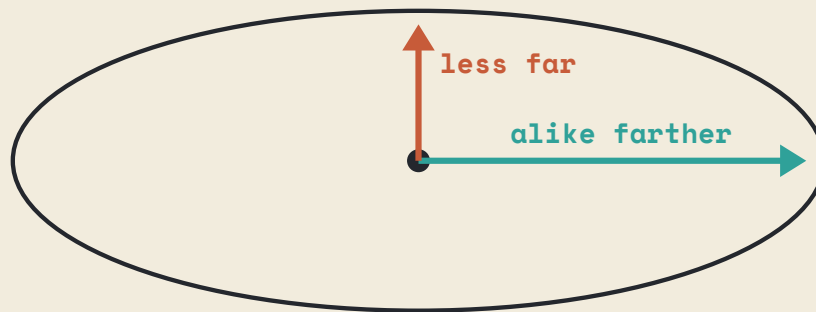
// the variogram $\gamma(h)$ --- nugget at the start, climbing to the sill at the range

FIG.15 --- A CHOICE OF CURVE



**Many smooth lines fit the dots.
We choose one -- that's our call.**

// the variogram model is a choice --- nugget, range and shape are decisions, not facts



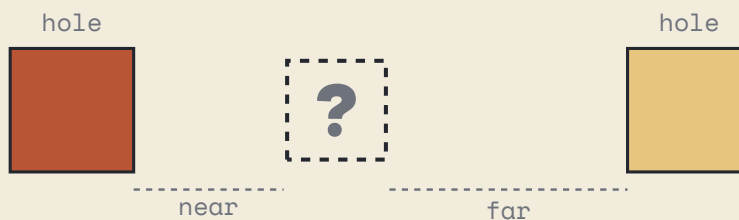
**Follow the grain and rocks
stay alike farther.**

// anisotropy --- the range depends on direction

ACT III

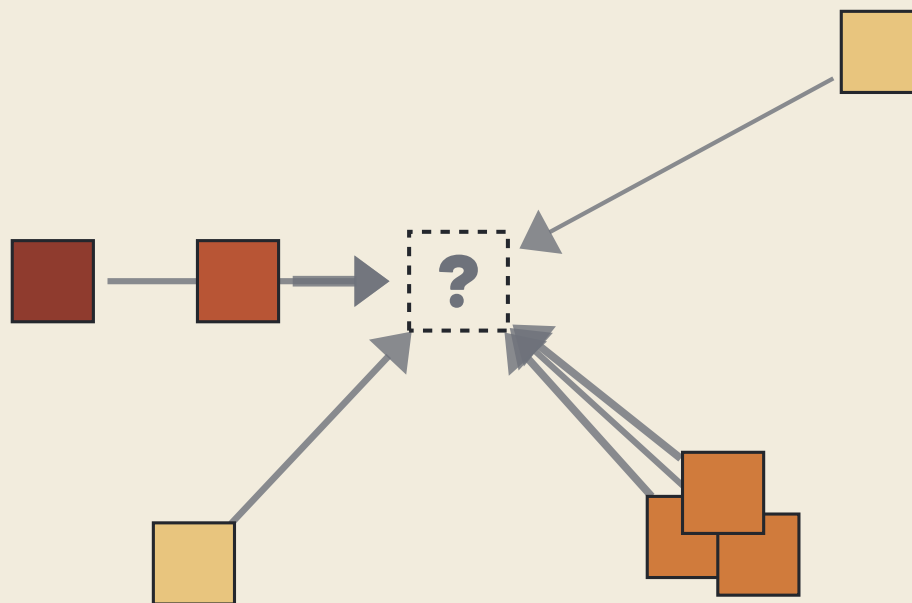
THE GUESS

filling the gaps, the fairest way --- kriging



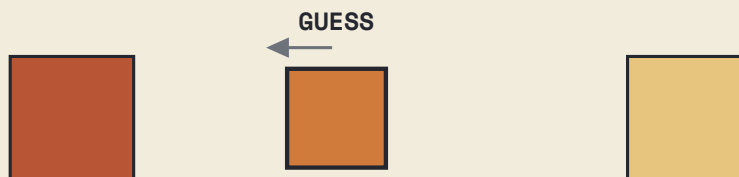
Here's a block we didn't drill.
What's inside?

// estimation --- predicting a value where we have no sample



Ask the rocks around it.
The closer ones speak louder.

// distance weighting --- nearer samples carry more weight

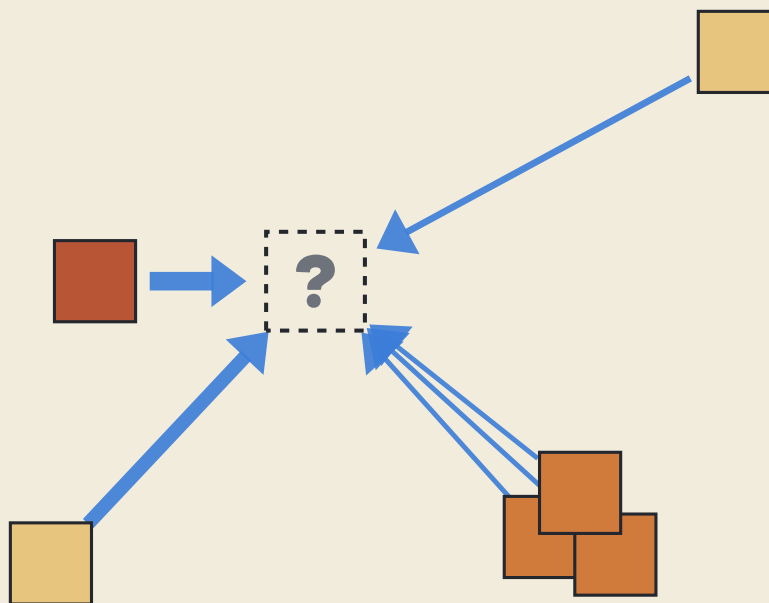


Mix the votes.
The guess leans to the nearer hole.

// a weighted average --- the estimate at the block

$\gamma(h)$

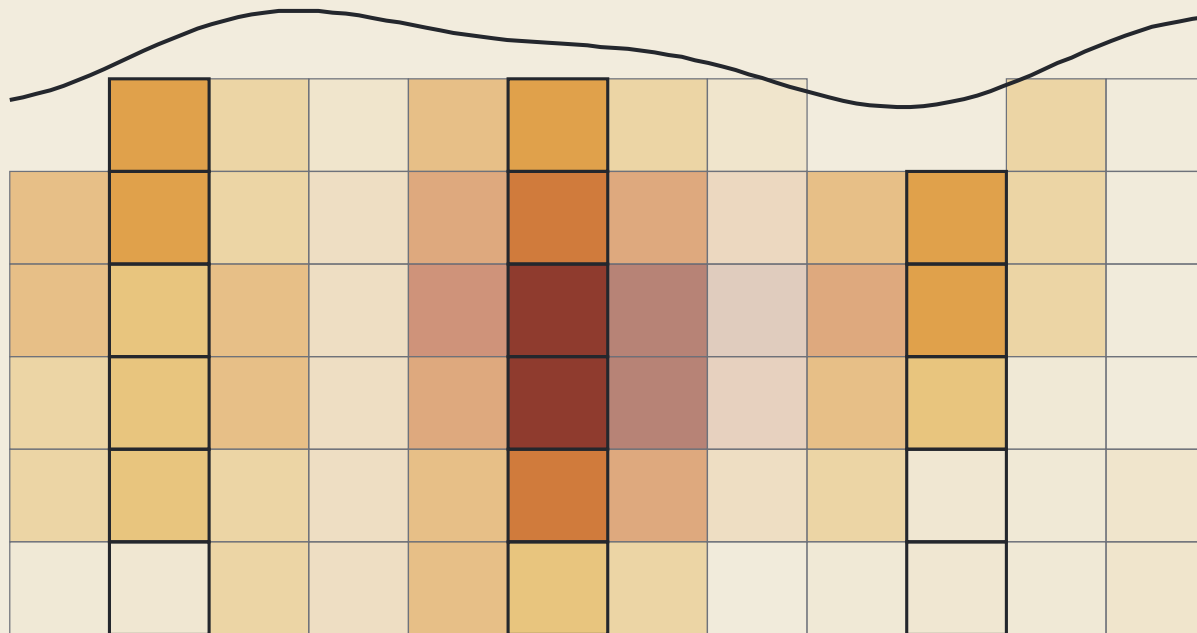
shares the votes



**Use the difference curve
to share the votes just right.**

// kriging --- the best linear unbiased estimate, weights from $\gamma(h)$

FIG.21 --- HOW SURE



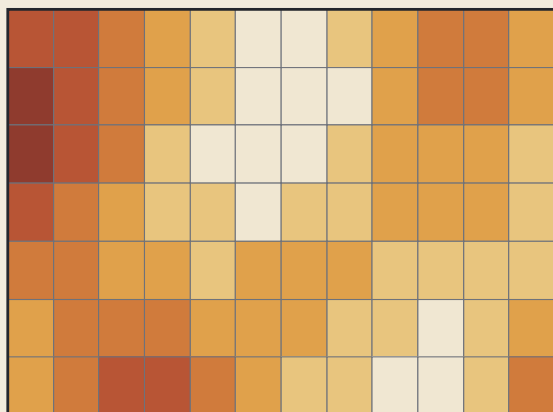
Near holes: sure. Far away: fuzzy.
Every block gets a fuzziness.

// the kriging variance --- an uncertainty for each estimate

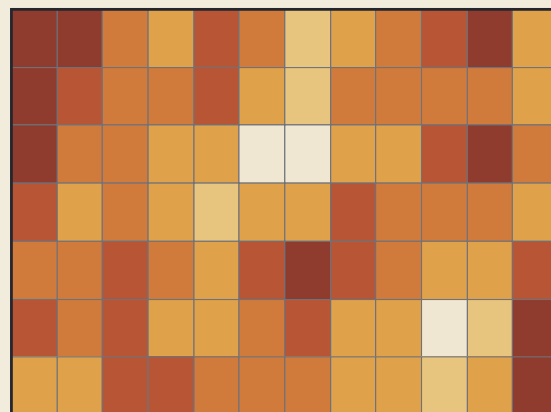
ACT IV

MANY GROUNDS

why one guess isn't enough --- simulation



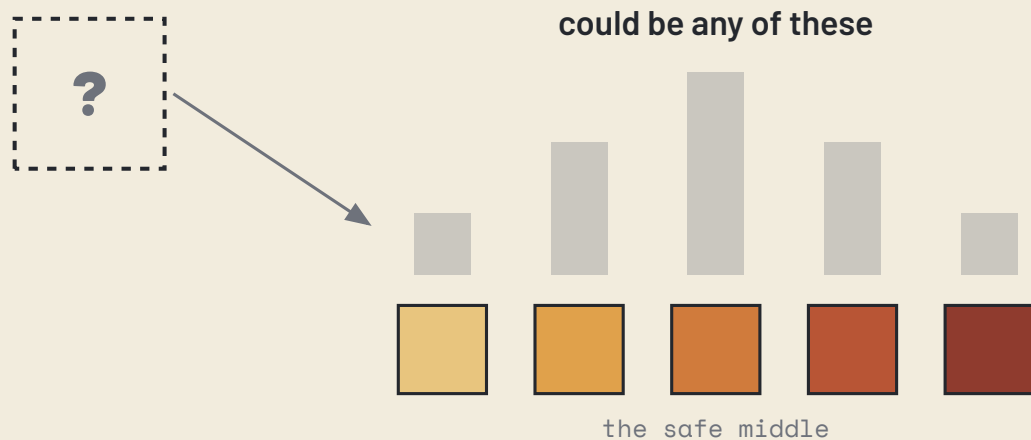
OUR GUESS



REAL ROCK

**Our one best guess is
gentle and blurry. Real rock is rougher.**

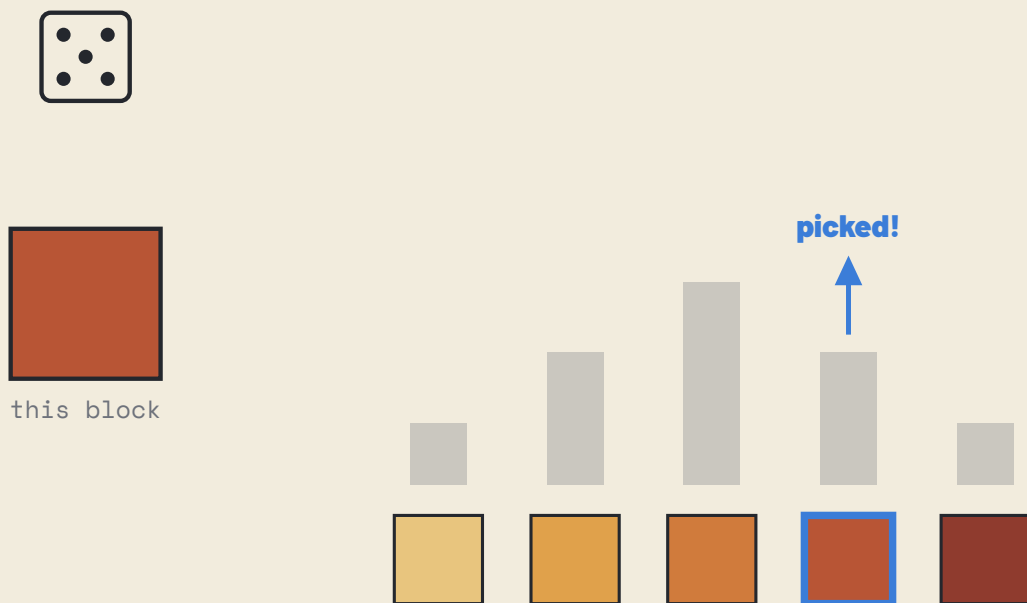
// kriging smooths --- because it takes the safe middle of every block's range



**At each block the guess isn't one colour --
it's a range of maybes.**

// the conditional distribution: centred on the kriged guess, as wide as the kriging fuzziness

FIG.24 --- ROLL THE DICE



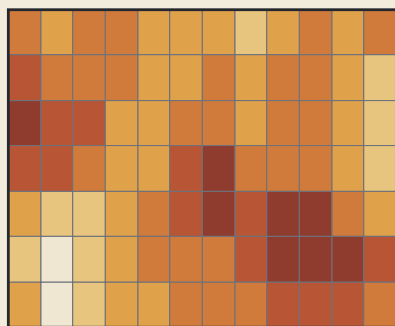
**Don't always take the safe middle.
Roll the dice -- pick one.**

```
// simulation draws a value from that bell instead of its mean --- that is the roughness coming  
back
```

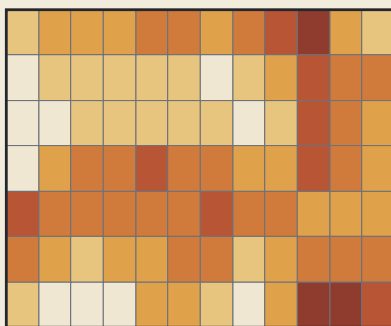


Your pick counts now.
On to the next block -- and the next.

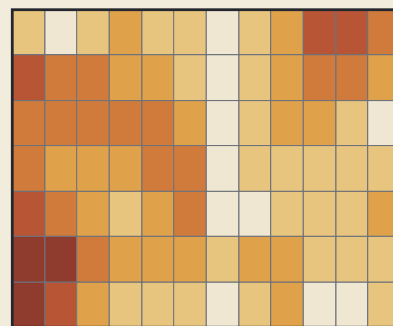
// sequential simulation: each drawn value becomes data for the next, preserving the texture



hill 1



hill 2

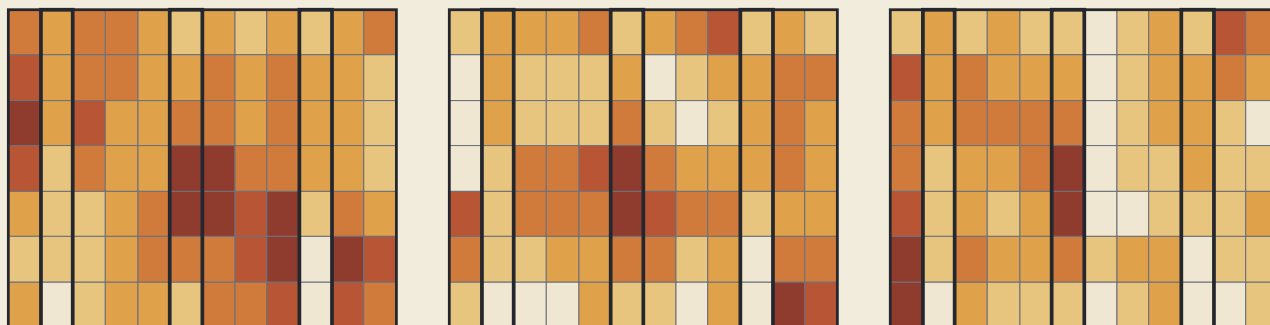


hill 3

Real rock is rough.
So we dream up lots of possible hills.

// geostatistical simulation --- many equally-likely realizations

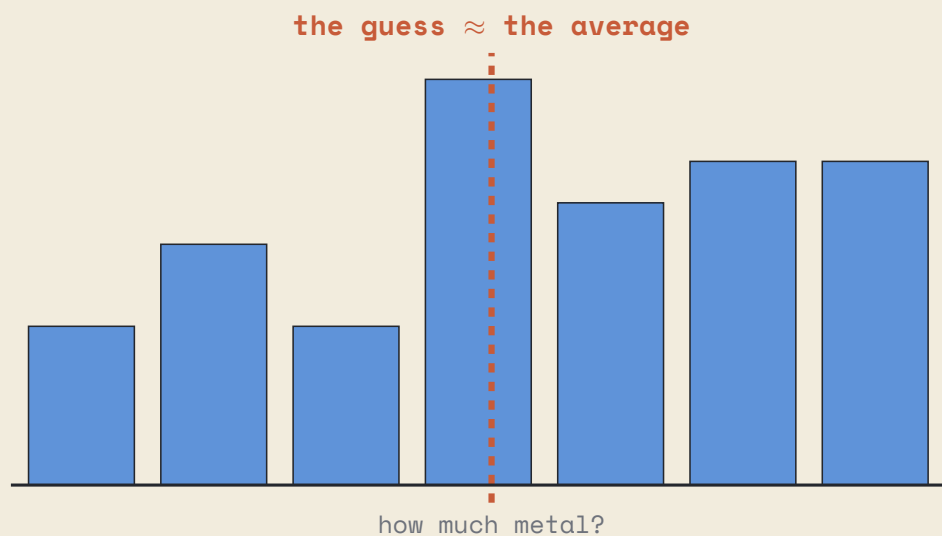
FIG.27 --- ALL FIT THE HOLES



**Every dreamed hill
still matches the holes we drilled.**

// conditional simulation --- realizations honour the data exactly

FIG.28 --- A SPREAD OF ANSWERS



Many hills → many answers.
Now we know our range.

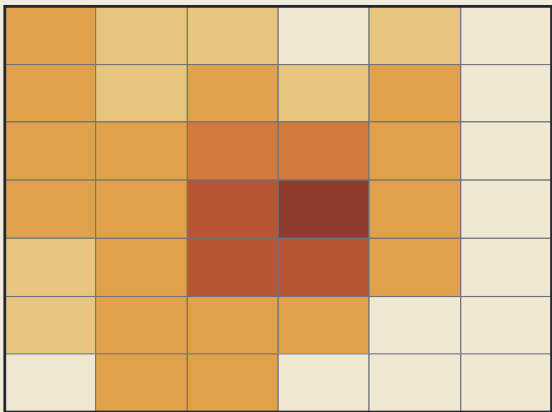
// uncertainty from realizations --- their average tracks the kriged estimate

ACT V

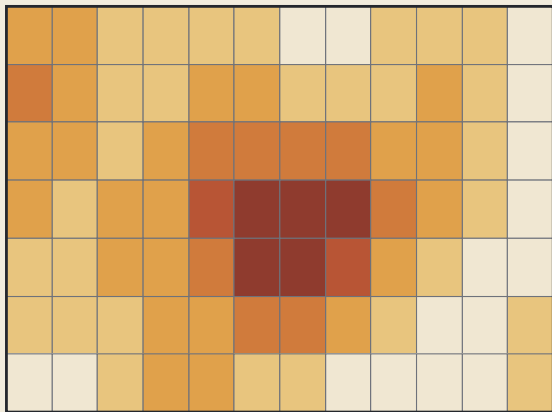
BLOCKS & DECISIONS

support, cutoffs, checking, and where to dig

FIG.29 --- BIG CALM, SMALL WILD



BIG BLOCKS --- calm

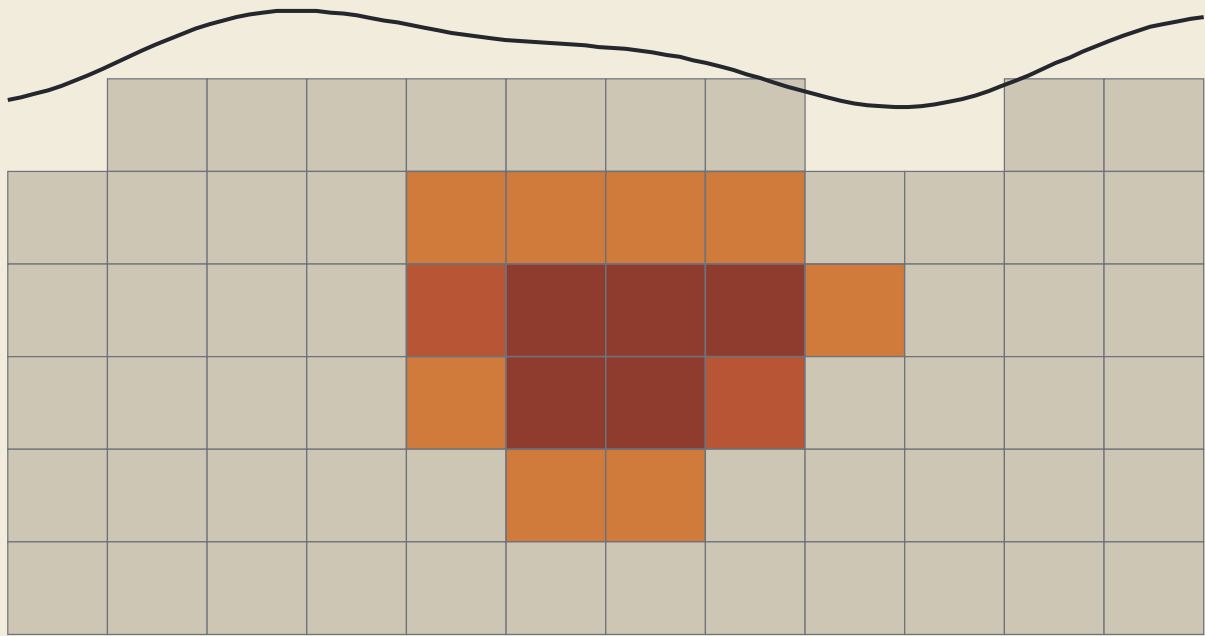


TINY BLOCKS --- wild

Big blocks are calm.
Tiny blocks are wild.

// support / the volume--variance effect --- larger volumes vary less

FIG.30 --- PICK A LINE

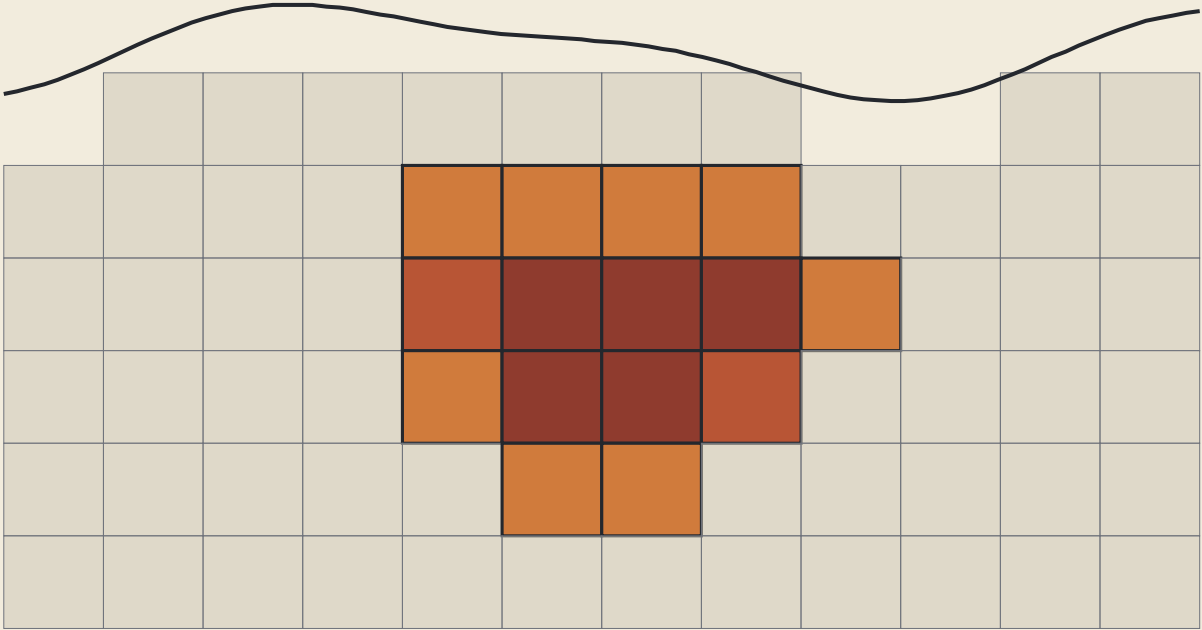


rich = dig · pale = leave

Rich enough to dig?
Yes, or no.

// cutoff grade --- an indicator splits ore from waste

FIG. 31 --- WORTH IT

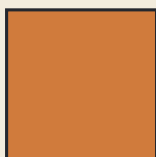


15 blocks worth digging

**Count the blocks
worth digging.**

// grade--tonnage --- how much rock sits above the cutoff

TRUE (hidden)



OUR GUESS

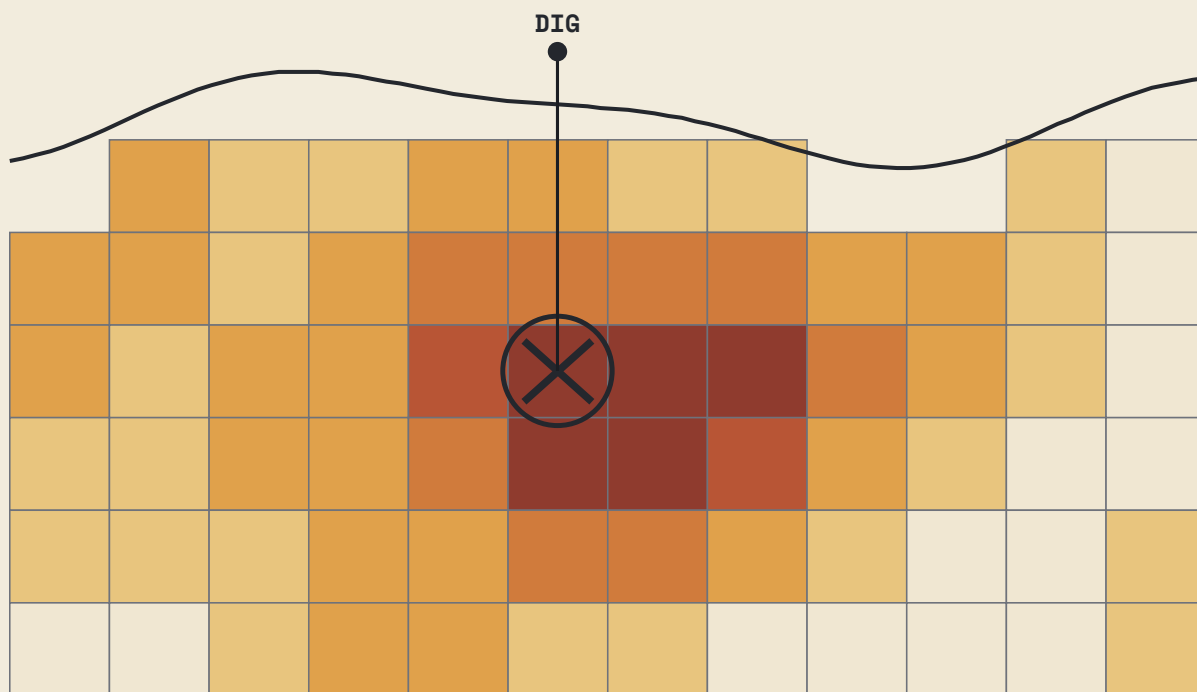


close!

Hide a hole. Guess it.
Were we close?

// cross-validation --- leave one out, predict it, measure the error

FIG.33 --- WHERE TO DIG



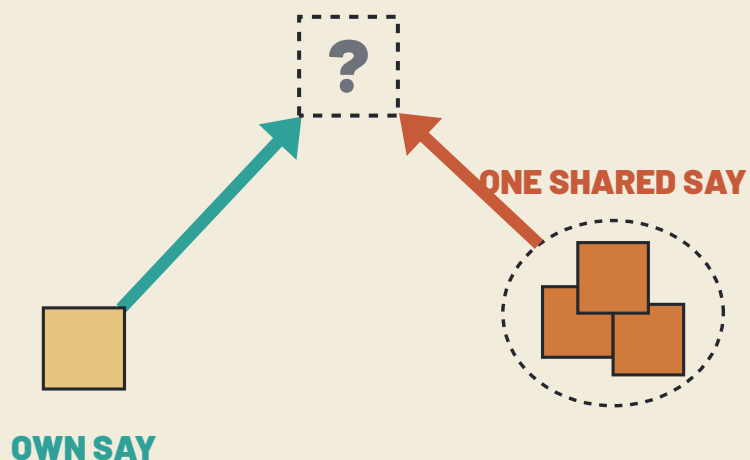
**Now we know the best spot --
and how sure we are.**

// the decision --- a block model turned into a plan

ACT VI

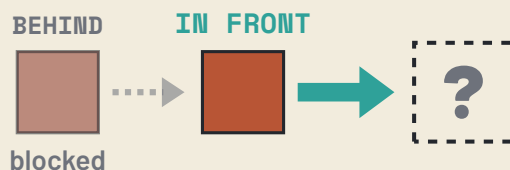
BONUS

for the grown-up reading along --- two kriging quirks



**A huddle shares one say.
The lonely rock isn't drowned out.**

// declustering --- kriging splits redundant clustered weight



**A rock in front
blocks the one behind it.**

// the screen effect --- a shadowed sample's weight collapses (it can even go negative)

FIELD KIT · No. 01

GLOSSARY

the real words, for the grown-up

regionalized variable

a value that depends on where you are

sample

one measured peek at the ground

core

the rock pulled up along a drillhole

stationarity

assuming the rules of variation hold everywhere

spatial correlation

near things tend to be alike

variogram

the curve of difference versus distance

nugget

variability left even at zero distance

range

the distance beyond which there's no correlation

sill

the total variability, reached at the range

anisotropy

correlation reaching farther in some directions

kriging

the best weighted estimate, using the variogram

kriging variance

the uncertainty attached to each estimate

support

bigger volumes vary less (volume--variance)

simulation

many possible grounds that fit the data

realization

one such possible ground

declustering

sharing the weight of bunched-up samples

screen effect

a near sample shadowing a farther one

cutoff grade

the line between ore and waste

cross-validation

hide a sample, predict it, check the error

capping / top-cut

taming an extreme outlier value

trend

a smooth, large-scale drift in grade

FIELD KIT No. 01

TO LEARN MORE

the grown-up's shelf --- in chronological order

W. Tobler (1970). A computer movie simulating urban growth in the Detroit region. *Economic Geography* 46: 234--240.

G. Matheron (1971). *The Theory of Regionalized Variables and Its Applications*. Cahiers du CMM, Fontainebleau.

A. Journel & C. Huijbregts (1978). *Mining Geostatistics*. Academic Press.

E. Isaaks & R. Srivastava (1989). *An Introduction to Applied Geostatistics*. Oxford Univ. Press.

N. Cressie (1993). *Statistics for Spatial Data*. Wiley.

P. Goovaerts (1997). *Geostatistics for Natural Resources Evaluation*. Oxford Univ. Press.

C. Deutsch & A. Journel (1998). *GSLIB: Geostatistical Software Library*. Oxford Univ. Press.

H. Wackernagel (2003). *Multivariate Geostatistics*, 3rd ed. Springer.

J.-P. Chilès & P. Delfiner (2012). *Geostatistics: Modeling Spatial Uncertainty*, 2nd ed. Wiley.

M. Rossi & C. Deutsch (2014). *Mineral Resource Estimation*. Springer.

A complete tour of geostatistics for the very young and the merely curious: we cannot see the ground, so we drill a few holes, measure how the rock hangs together, guess the rest the fairest way, and — because one guess is too tidy — dream up many possible grounds to learn how sure we really are.

How to read this book: the big line is to say out loud. The small // line names the real idea, for the grown-up.

FIELD KIT No.01 --- ``What's Under the Hill?''

Geoscientific Chaos Union · public domain (CC0) · ground simulated from seed 7

Barlow & Space Mono · drawn in TikZ · proof edition, not yet imposed for print